

2025 HSC Mathematics Standard 1 Marking Guidelines

Section I

Multiple-choice Answer Key

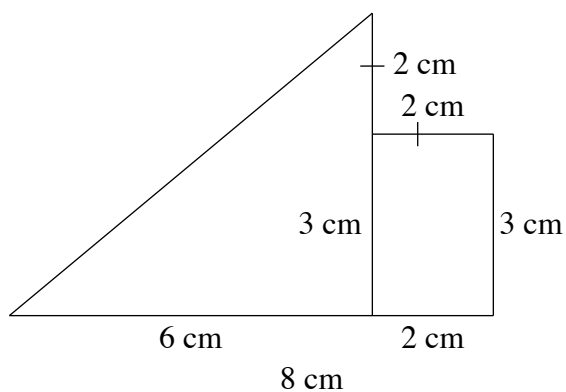
Question	Answer
1	D
2	B
3	B
4	C
5	A
6	C
7	C
8	A
9	D
10	B

Section II

Question 11

Criteria	Marks
• Provides correct answer	3
• Finds the area of the triangle, or equivalent merit	2
• Finds an area of a rectangle, or equivalent merit	1

Sample answer:



Area of rectangle: $3 \times 2 = 6 \text{ cm}^2$

Area of triangle: $\frac{1}{2} \times 6 \times 5 = 15 \text{ cm}^2$

Total area = $15 \text{ cm}^2 + 6 \text{ cm}^2 = 21 \text{ cm}^2$

Question 12

Criteria	Marks
• Provides correct answer	2
• Finds the difference between the heart rates, or equivalent merit	1

Sample answer:

$$\begin{aligned}\text{Increase} &= 126 - 72 \\ &= 54\end{aligned}$$

$$\begin{aligned}\% \text{ Increase} &= \frac{54}{72} \times 100 \\ &= 75\%\end{aligned}$$

Question 13 (a)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}&\frac{1}{2} \times h \times (a + b) \\ &= \frac{1}{2} \times 12 \times (a + 30) \\ &= 6(a + 30)\end{aligned}$$

Question 13 (b)

Criteria	Marks
• Provides correct solution	2
• Writes a correct equation with the given area, or equivalent merit	1

Sample answer:

$$\begin{aligned}330 &= 6(a + 30) \\ 55 &= a + 30 \\ 25 &= a \\ \therefore a &= 25\end{aligned}$$

Question 14 (a)

Criteria	Marks
• Provides correct answer	2
• Provides shortest path, or equivalent merit	1

Sample answer:

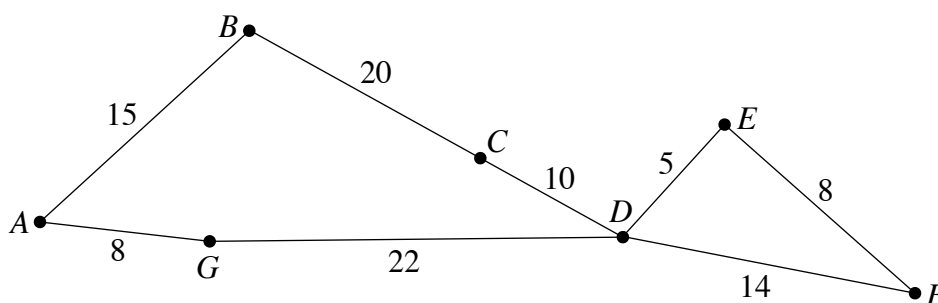
Minimum time travel = $15 + 20 + 10 + 5 + 8$
 = 58 minutes

Path: $ABCDEF$

Question 14 (b)

Criteria	Marks
• Provides correct diagram	2
• Connects A to D , through G , or equivalent merit	1

Sample answer:



Question 14 (c)

Criteria	Marks
• Provides correct reason	1

Sample answer:

The path in part (a) is not the shortest path any more as $AGDEF$ takes 43 minutes.

Question 15 (a)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Is there a relationship between average number of minutes per day watching television and average number of minutes per day exercising?

Question 15 (b)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Average minutes per day exercising

Question 15 (c)

Criteria	Marks
• Provides correct answer	2
• Provides correct direction or form	1

Sample answer:

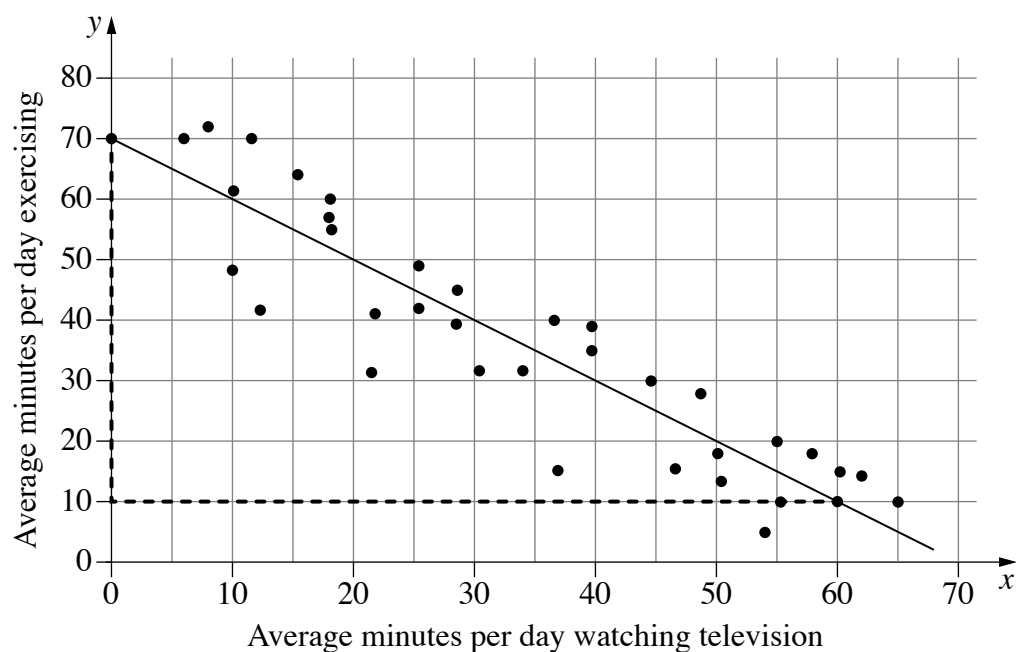
Form: Linear

Direction: Negative

Question 15 (d)

Criteria	Marks
• Provides correct solution	3
• Provides correct intercept of the line, or equivalent merit	2
• Correctly plots the given points on the graph	1

Sample answer:



$$m = \frac{-60}{60} \quad y\text{-intercept} = 70$$

$$= -1$$

Question 15 (e)

Criteria	Marks
• Provides correct reason	1

Sample answer:

The number of minutes per day exercising will be negative.

Question 16

Criteria	Marks
• Provides correct solution	3
• Finds the constant of variation, or equivalent merit	2
• Provides a correct direct variation statement, or equivalent merit	1

Sample answer:

$$M = kA$$

$$500 = k \times 5^2$$

$$20 = k$$

$$M = 20A$$

$$M = 20 \times 3^2$$

$$M = 180 \text{ g}$$

Question 17

Criteria	Marks
• Provides correct solution	2
• Finds the cost of ten-litre cans, or equivalent merit	1

Sample answer:

$$\begin{aligned} \text{Cost of ten-litre cans: } & \$205 \times 8 \\ & = \$1640 \end{aligned}$$

$$\begin{aligned} \text{Cost of four-litre cans: } & \$90 \times 20 \\ & = \$1800 \end{aligned}$$

$$\begin{aligned} \text{Savings} & = \$1800 - \$1640 \\ & = \$160 \end{aligned}$$

Question 18 (a)

Criteria	Marks
• Provides correct answer	2
• Converts 1.25 hours to 1 hour and 15 minutes, or equivalent merit	1

Sample answer:

1.25 hours = 1 hour 15 minutes

20 minutes + 35 minutes + 1 hour + 15 minutes

= 70 minutes + 1 hour

= 2 hours 10 minutes

Question 18 (b)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\frac{100 \text{ km}}{1.25 \text{ hours}} = 80 \text{ km/h}$$

Question 19 (a)

Criteria	Marks
• Provides correct solution	3
• Uses the 37 cents in a relevant calculation, or equivalent merit	2
• Identifies the correct row in the tax table, or equivalent merit	1

Sample answer:

$$\begin{aligned} \text{Tax payable} &= 31\,288 + 0.37(148\,600 - 135\,000) \\ &= \$36\,320 \end{aligned}$$

Question 19 (b)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$2\% \times 148\,600 = \$2972$$

Question 20 (a)

Criteria	Marks
• Provides correct answer	2
• Attempts to use diagram and scale	1

Sample answer:

$$15 \times 5 = 75 \text{ m} \quad \text{OR} \quad 7.5 \times 10 = 75 \text{ m}$$

Question 20 (b)

Criteria	Marks
• Provides correct solution	3
• Finds the length of AC , or equivalent merit	2
• Attempts to use Pythagoras' theorem, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \text{Length of } AC &= \sqrt{75^2 + 75^2} \\
 &= 106.066... \\
 &= 106 \text{ m (correct to 3 significant figures)}
 \end{aligned}$$

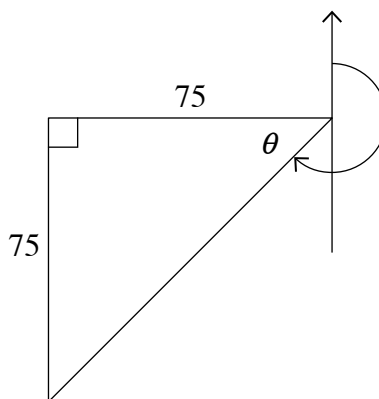
Question 20 (c)

Criteria	Marks
• Provides correct answer	2
• Finds the size of angle BCA , or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \tan \theta &= \frac{75}{75} \\
 \theta &= 45^\circ
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing is } 180^\circ + (90^\circ - 45^\circ) \\
 &= 225^\circ
 \end{aligned}$$



Question 21

Criteria	Marks
• Provides correct solution	3
• Uses the compound interest formula with correct r , or equivalent merit	2
• Attempts to use compound interest formula, or equivalent merit	1

Sample answer:

$$\text{Interest days} = 68 - 45 = 23$$

$$\begin{aligned}\text{Amount} &= \$392 \left(1 + \frac{13.74}{365} \% \right)^{23} \\ &= \$395.41\end{aligned}$$

$$\begin{aligned}\text{Interest charged} &= \$395.41 - \$392 \\ &= \$3.41\end{aligned}$$

Question 22 (a)

Criteria	Marks
• Provides correct solution	2
• Uses 2.4 in Pythagoras' theorem, or equivalent merit	1

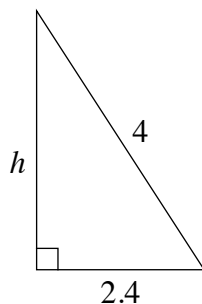
Sample answer:

$$h^2 = 4^2 - 2.4^2$$

$$h^2 = 10.24$$

$$h = \sqrt{10.24}$$

$$h = 3.2 \text{ cm}$$



Question 22 (b)

Criteria	Marks
• Provides correct solution	2
• Finds the area of the circle, or equivalent merit	1

Sample answer:

$$\text{Area of triangle} = 7.68$$

$$\begin{aligned} \text{Area of circle} &= \pi \times 2.5^2 \\ &= 19.635 \end{aligned}$$

$$\frac{7.68}{19.635} \times 100\% = 39.1\%$$

Question 23

Criteria	Marks
• Provides correct answer	2
• Provides one correct linear equation	1

Sample answer:

$$P = 11B$$

$$\$465P + \$350B = \$494\,000\,000$$

Question 24

Criteria	Marks
• Provides correct solution	5
• Provides total amount to be repaid, or equivalent merit	4
• Calculates the interest on the loan, or equivalent merit	3
• Finds the total amount to be borrowed, or equivalent merit	2
• Finds the number of \$100 needed to calculate stamp duty, or equivalent merit	1

Sample answer:

$$\begin{aligned}\text{Stamp duty} &= \$24\,200 \div 100 \\ &= 242\end{aligned}$$

$$\begin{aligned}\therefore \text{Total stamp duty} &= 242 \times 3 \\ &= \$726\end{aligned}$$

$$\begin{aligned}\text{Amount borrowed} &= \$726 + \$24\,200 + \$50 \\ &= \$24\,976\end{aligned}$$

$$\begin{aligned}\text{Interest on loan} &= \$24\,976 \times 6.8\% \times 3 \\ &= \$5095.10\end{aligned}$$

$$\begin{aligned}\text{Total amount to be repaid} &= \$24\,976 + \$5095.10 \\ &= \$30\,071.10\end{aligned}$$

$$\begin{aligned}\text{Monthly repayment} &= \$30\,071.10 \div 36 \\ &= \$835.31\end{aligned}$$

Question 25

Criteria	Marks
• Provides correct answer	4
• Finds the correct return for one investment, or equivalent merit	3
• Attempts to use the compound interest formula, or equivalent merit	2
• Attempts to find the simple interest, or equivalent merit	1

Sample answer:

$$25\,000 \left(1 + \frac{0.05}{12}\right)^{120} = 41\,175.24 \quad \text{for Option A}$$

$$25\,000 + 25\,000 \times 0.08 \times 10 = 45\,000 \quad \text{for Option B}$$

$\therefore$ Bobbie should choose option B.

Question 26

Criteria	Marks
• Provides correct solution	4
• Calculates the cost of tiles, or equivalent merit	3
• Calculates the area of the rectangle in m^2 , or equivalent merit	2
• Attempts to use the scale, or equivalent merit	1

Sample answer:

10.4 cm by 9.6 cm

$$10.4 \times 50 \text{ cm} = 520 \text{ cm} = 5.2 \text{ m}$$

$$9.6 \times 50 \text{ cm} = 480 \text{ cm} = 4.8 \text{ m}$$

$$\text{Area} = 5.2 \text{ m} \times 4.8 \text{ m} = 24.96 \text{ m}^2$$

$$\text{Total area of tiles required} = 24.96 + 15\% \text{ of } 24.96 = 28.704$$

$$\$150 \times 28.704 = \$4305.60$$

$$\text{Tiler is } \$90 \times 20 = \$1800$$

$$\text{Total cost is } \$4305.60 + \$1800 = \$6105.60$$

\$6106 to nearest dollar

Answers could include:

Total area of tiles required could be rounded to 29 m^2 .

Question 27

Criteria	Marks
• Provides correct solution	4
• Finds value after 9 years, or equivalent merit	3
• Provides correct rate, or equivalent merit	2
• Uses 44 000 correctly in an appropriate formula, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \$44\,000 &= \$55\,000(1 - r) & \text{OR} & \quad 55\,000 - 44\,000 = 11\,000 \\
 \frac{44\,000}{55\,000} &= 1 - r & & \quad r = \frac{11\,000}{55\,000} \times 100\% = 20\% \\
 0.8 &= 1 - r \\
 \therefore r &= 0.2 \\
 r &= 20\%
 \end{aligned}$$

$$\begin{aligned}
 \text{Value after 9 years} & & \text{OR} & & \text{Value after 9 years} \\
 = \$55\,000(1 - 0.2)^9 & & & & = \$55\,000(1 - 0.2)^9 \\
 = \$7381.98 & & & & = \$7381.98 \\
 \\
 \text{Value of depreciation} & & & & \text{Value after 10 years} \\
 = \$7381.98 \times 0.2 & & & & = \$55\,000(1 - 0.2)^{10} \\
 = \$1476.40 & & & & = \$5905.58 \\
 \\
 & & & & \text{Difference} = \$1476.40
 \end{aligned}$$

Question 28 (a)

Criteria	Marks
• Provides correct solution	2
• Finds the mass of one coin in kg, or equivalent merit	1

Sample answer:

500 \$2 coins have a mass of 3.3 kg

Therefore, the mass of 1 \$2 coin is $\frac{3.3 \text{ kg}}{500} = 0.0066 \text{ kg}$

$0.0066 \times 1000 = 6.6 \text{ grams}$

Question 28 (b)

Criteria	Marks
• Provides correct solution	2
• Finds the mass of one note, or equivalent merit	1

Sample answer:

200 notes have a mass of 157 g

Therefore, the mass of one note is $\frac{157 \text{ g}}{200} = 0.785 \text{ g}$

The mass of 500 notes is $0.785 \times 500 = 392.5 \text{ g}$
 $= 393 \text{ g}$ (to the nearest gram)

2025 HSC Mathematics Standard 1 Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	MS-S1 Data Analysis	MS11-10
2	1	MS-M4 Rates	MS1-12-3
3	1	MS-N1 Networks and Paths	MS1-12-8
4	1	MS-M1 Applications of Measurement	MS11-3
5	1	MS-A3 Types of Relationships	MS1-12-1
6	1	MS-N1 Networks and Paths	MS1-12-8
7	1	MS-S2 Relative Frequency and Probability	MS11-10
8	1	MS-S2 Relative Frequency and Probability	MS11-10
9	1	MS-F2 Investment	MS1-12-5
10	1	MS-M5 Scale Drawing	MS1-12-3

Section II

Question	Marks	Content	Syllabus outcomes
11	3	MS-M1 Applications of Measurement	MS11-4
12	2	MS-M4 Rates	MS1-12-3
13 (a)	1	MS-M1 Applications of Measurement	MS11-4
13 (b)	2	MS-A1 Formulae and Equations	MS11-1
14 (a)	2	MS-N1 Networks and Paths	MS1-12-8
14 (b)	2	MS-N1 Networks and Paths	MS1-12-8
14 (c)	1	MS-N1 Networks and Paths	MS1-12-8, MS1-12-10
15 (a)	1	MS-S3 Further Statistical Analysis	MS1-12-2
15 (b)	1	MS-S3 Further Statistical Analysis	MS1-12-2
15 (c)	2	MS-S3 Further Statistical Analysis	MS1-12-2
15 (d)	3	MS-S3 Further Statistical Analysis	MS1-12-7
15 (e)	1	MS-S3 Further Statistical Analysis	MS1-12-2, MS1-12-10
16	3	MS-A2 Linear Relationships	MS11-10
17	2	MS-M4 Rates	MS1-12-10
18 (a)	2	MS-M2 Working with Time	MS1-11-3
18 (b)	1	MS-M4 Rates	MS1-12-3
19 (a)	3	MS-F1 Money Matters	MS11-5
19 (b)	1	MS-F1 Money Matters	MS11-5
20 (a)	2	MS-M5 Scale Drawing	MS1-12-4
20 (b)	3	MS-M3 Right-angled Triangles	MS1-12-3
20 (c)	2	MS-M3 Right-angled Triangles	MS1-12-4
21	3	MS-F3 Depreciation and Loans	MS1-12-5, MS1-12-10
22 (a)	2	MS-M3 Right-angled Triangles	MS1-12-3

Question	Marks	Content	Syllabus outcomes
22 (b)	2	MS-M1 Applications of Measurement	MS11-4
23	2	MS-A3 Types of Relationships	MS1-12-6
24	5	MS-F1 Money Matters	MS11-5, MS11-10
25	4	MS-F2 Investment	MS1-12-5, MS1-12-10
26	4	MS-M5 Scale Drawing	MS1-12-3
27	4	MS-F3 Depreciation and Loans	MS1-12-5
28 (a)	2	MS-M1 Applications of Measurement MS-M4 Rates	MS1-12-3
28 (b)	2	MS-M1 Applications of Measurement MS-M4 Rates	MS1-12-3, MS1-12-10